

Question 1:

Object:

Create a PAC (Program Analysis Chart) for a program that calculates the total cost of a meal including tax (as a percentage) and tip. Display the final cost after tax and tip.

PROGRAM ANALYSIS CHART

<u>Given Data</u>	<u>Expected Outcome</u>
Meal cost Tax percentage Tip percentage	Final total cost
<u>Process Required</u>	<u>Situation Alternatives</u>
tax amount = (meal cost × tax%) ÷ 100. tip amount = (meal cost × tip%) ÷ 100. final cost = meal cost + tax amount + tip amount.	If tax = 0 → Final = meal + tip If tip = 0 → Final = meal + tax If both tax & tip = 0 → Final = meal only If meal cost ≤ 0 → Invalid input

Question 2:

Object:

Draw an IPO (Input-Process-Output) chart for a program that calculates the area of a circle.

Formula: $\text{Area} = \pi * \text{radius}^2$

INPUT PROCESSING OUTPUT CHART

<u>Input</u>	<u>Process</u>	<u>Module Reference</u>	<u>Output</u>
Radius of circle	$\text{Area} = \pi \times \text{radius}^2$	READ Radius CALCULATE $\text{Area} = \pi \times \text{radius}^2$ PRINT Area	Area of the circle

Question 3:

Object:

Write a step-by-step algorithm to calculate the body mass index (BMI) of a person.

Formula: $BMI = \text{weight (kg)} / (\text{height (m)})^2$.

ALGORITHM

Step 01: Start the Program.

Step 02: Declare variables Name, Weight, Height, BMI.

Step 03: Read Name, Weight (in kilograms), Height (in meters).

Step 04: Calculate $BMI = Weight / (Height \times Height)$.

Step 05: Write: "Hi {Name}, your BMI is {BMI}."

Step 06: Stop the Program.

Question 4:

Object:

Write pseudocode for a program that calculates the distance traveled by a car and display it.

Calculate the distance using the Formula: $\text{Distance} = \text{Speed} * \text{Time}$.

PSEUDO CODE

STARAT

INPUT Speed

INPUT Time

SET Distance = Speed * Time

OUTPUT "The Distance is: " + Distance

STOP

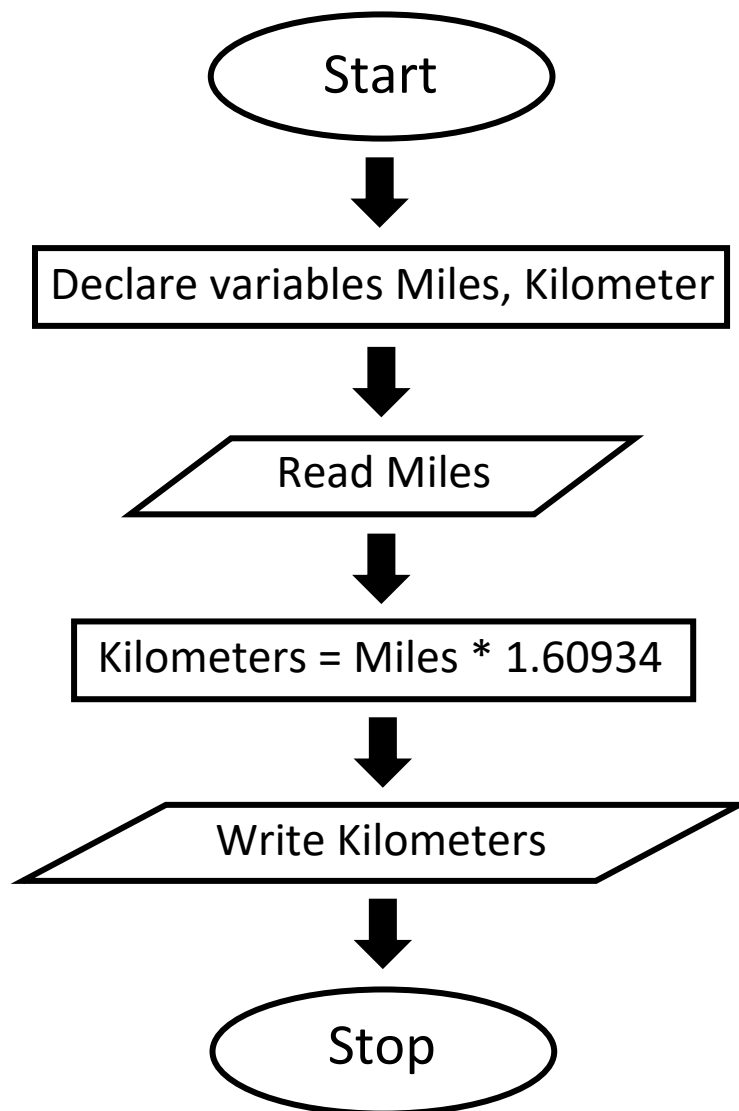
Question 5:

Object:

Draw a flowchart that represents the process of converting a distance from miles to kilometers.

The flowchart should start with receiving the distance in miles as input. It should then process the calculation ($\text{kilometers} = \text{miles} * 1.60934$) and finish with displaying the output (the distance in kilometers). Use standard flowchart symbols.

FLOWCHART



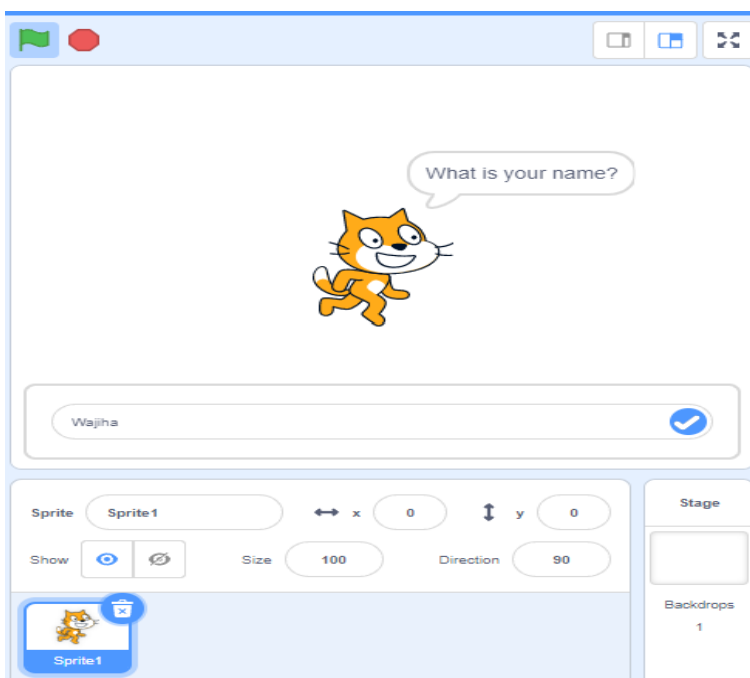
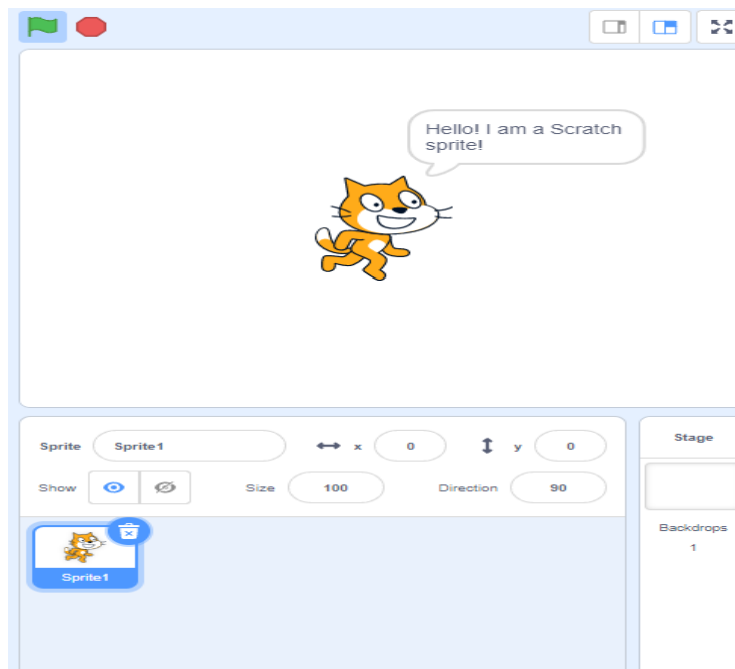
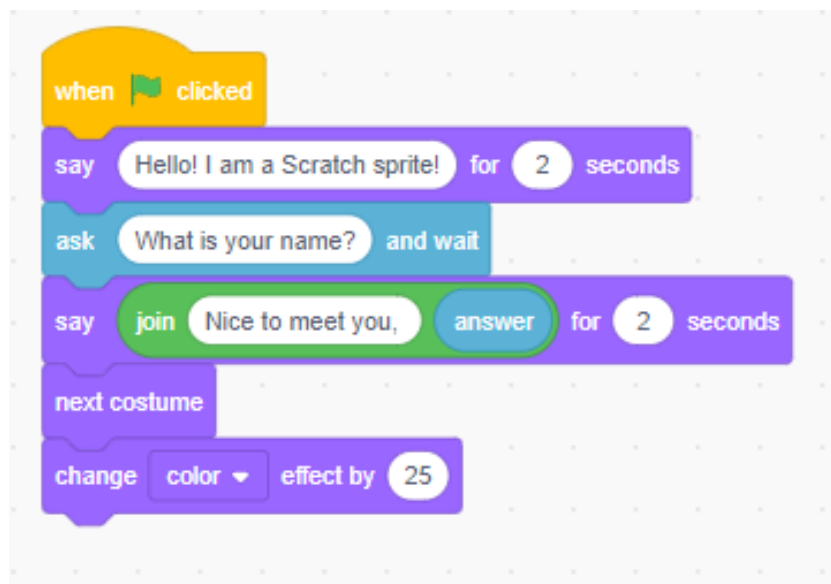
Question:6

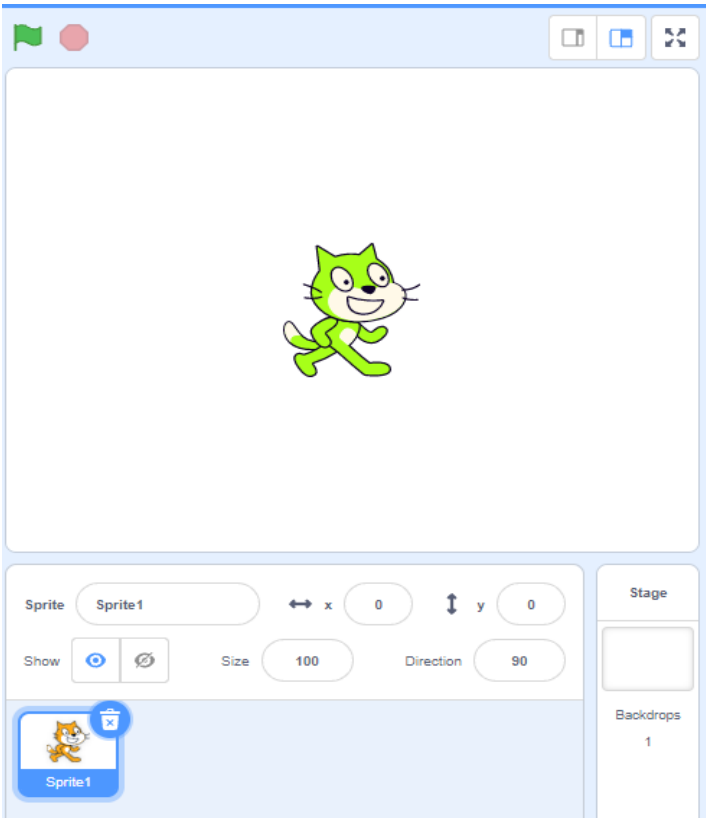
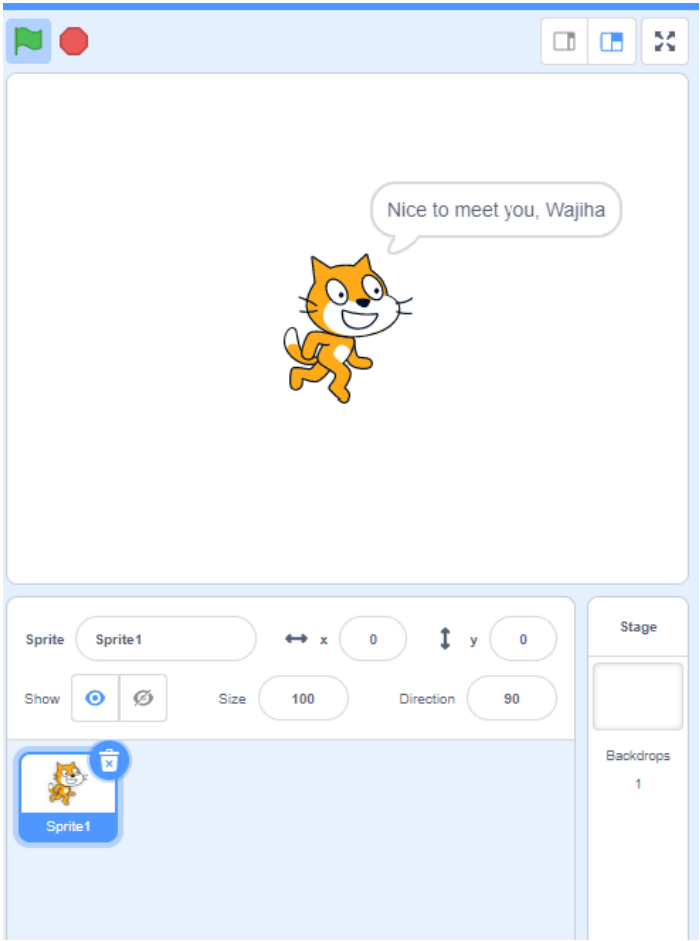
Object:

Create a Scratch program where a sprite introduces itself and changes its appearance.

1. When the green flag is clicked, the sprite says “Hello! I am a Scratch sprite!” for 2 seconds.
2. The sprite then asks “What is your name?” and waits for the user to type an answer.
3. After getting the answer, the sprite says “Nice to meet you, [answer]!” for 2 seconds.
4. Finally, the sprite changes its costume to a different pre-existing costume and changes its color effect.

SCRACH





Question:7

Object:

Create a Scratch program where a sprite moves around the stage in a simple pattern. When the green flag is clicked, the sprite goes to the center of the stage (x:0, y:0).

The sprite then glides to four different positions on the stage one after another:

- First, glide to the top (x:0, y:100)
- Second, glide to the right (x:100, y:0)
- Third, glide to the bottom (x:0, y:-100)
- Fourth, glide to the left (x:-100, y:0)

Each time the sprite reaches a new position, it should say “Hello!” for 1 second. After visiting all four positions, the sprite returns to the center and says “That was fun!”.

SCRACH

